

Final Report: Evaluating the Inflationary Impact of Tariffs on USA

Introduction

Our goal is to utilize historical tariff and CPI data for the past 10 years to analyze how changes in tariffs on imports into the USA affect consumer prices in three key industries – electronics, food, and apparel. By visualizing the relationship between tariff rates and inflation through an interactive user dashboard, we seek to make the inflationary effect of tariff increases understandable to stakeholders such as consumers, businesses and lawmakers.

Problem Definition

There is no single, interactive tool enabling users to dynamically explore tariff pass-through, supply chain responses, and consumer price implications in one accessible dashboard. This is why we proposed this project, to build an interactive tool for general users to help them understand and draw insights about tariff changes and inflationary responses without the need of experts.

Literature Survey

Current research on tariffs and their effects relies primarily on econometric and macroeconomic modeling, static welfare analysis, and sector focused empirical estimation. There are existing studies that quantify the short-term consumer and welfare losses caused by tariff impositions, demonstrating that higher import taxes directly increase retail prices and reduce consumer welfare [1], [2]. Other studies extend this work by introducing uncertainty and dynamic modeling to capture delayed and indirect inflationary responses [3], [4]. Retail and supply chain behavior is also an active area of study, where works examine firm-level markup adjustments and heterogeneous pass-through effects, revealing that price impacts vary significantly by product type and industry [5]. Further evidence highlights that tariff effects differ by firm type, industry, and product uses [6]. However, such studies are typically constrained to narrow categories or specific trade disputes (such as US – China tariffs) [6], [7]. Broader analysis shows how tariffs disrupt supply networks and affect firm efficiency, investment, and financial risk, but they often stop short of linking these outcomes to real consumer level data [8], [9], [10]. Recent studies have begun incorporating real-time or micro-level data [11], [12] to identify tariff impacts as they emerge, while international [13] and developing country analysis [14] provide comparative insights but lack direct policy interactivity. Finally, there are structural modeling works that offer welfare frameworks [15] but remain highly theoretical and need specific skills to implement and interpret [5]. Overall, today's methods are fragmented across macroeconomic, financial, and retail perspectives, with limited integration for stakeholders to visualize or simulate policy effects.

Proposed Method

Our goal is to utilize historical tariff and CPI data for the past 10 years to analyze how changes in tariffs on imports into the USA affect consumer prices in three key industries – electronics, food, and apparel.

Innovations

This approach is innovative through the application of machine learning to industry-specific scenarios across key sectors for businesses, consumers, and policymakers. The interactive Tableau dashboard allows users to see, in real time, the impacts on consumer prices under different tariff scenarios, enabling customizable “what-if” analysis relevant to a wide range of stakeholders.

Forecast of CPI for the future years - we have used Exponential Smoothing (ETS) model - Holt-Winters exponential smoothing which is designed to project future values of a time series based on its past level, trend, and seasonality. In addition, an ARIMAX model has been developed to incorporate tariff rates as external variables, providing a more causally interpretable link between tariff changes and CPI fluctuations.

Simulation - Dynamic “what-if” scenario analysis: Allows users to manipulate tariff rates and instantly observe predicted inflationary responses using the ARIMAX model shown in Equation 1.

Industry-level focus - Enables targeted insights for electronics, food, and apparel rather than aggregate CPI.

Approach and Implementation

Data Collection and Cleaning:

- a. Data collected – Historical CPI data and tariff data with HTS codes over a 10-year period [16]
- b. Specific keywords were used to assign an industry. For example, keywords such as “meat”, “fruit” and/or “vegetable” were used to classify tariff rates to the ‘food’ industry.
- c. Handled missing CPI values via linear interpolation.
- d. Merged multiple datasets by year, country, and HS industry code.
- e. Conducted outlier detection and removal using interquartile range

Exploratory Data Analysis (EDA):

Using the base library of R, a correlation table was generated to examine the relationships between the CPI and PPI data across the selected industries, as well as corresponding tariff data. The table showed strong correlations between the tariff data and CPI/PPI data, which gave us confidence that trends in tariffs are linked to price index movements. This justified the use of an ARIMAX model, using the forecast library in R, to forecast future CPI values based on past tariff data. A key assumption of the ARIMAX model is that the underlying time series should be stationary. To assess this, we used autocorrelation (ACF) plots and Augmented Dickey–Fuller (ADF) tests on each series. The results showed that all the series were non-stationary, so first order differencing was applied to remove trends, and the tests were repeated. However, some of the data sets still exhibited non-stationarity and thus additional measures such as log transformation before the differencing were needed before delving into the ARIMAX model.

Dashboard Design:

Tableau was used to integrate tariff, CPI, and PPI datasets through date-based joins, enabling consistent trend analysis across industries. The dashboard combines high-level visuals, interactive elements, and model-generated forecasts to explore historical patterns and simulate tariff impacts. This resulted in a clear, intuitive tool that helps users understand how U.S. tariffs influence prices over time.

The final version of our dashboard can be seen in Figure 1.

Model Building and Simulation Framework:

Initial Model (Exponential Smoothing (ETS) model): Holt-Winters exponential smoothing, which is designed to project future values of a time series based on its past level, trend, and seasonality. The ETS model helps us capture patterns in the CPI data. By applying exponential weighting, the model then emphasizes the more recent data while also accounting for long-term movements. This served as a baseline to compare simulated tariff impacts across the industries of focus. We decided on this model since it provides an interpretable way to reliably forecast changes in CPI and tariff rates.

Tariff Forecasting (ETS Model): The historical tariff data is modeled using an Exponential Smoothing (ETS) model to forecast expected tariff rates for the short-term future. This model was chosen since tariff data often changes over time with policy-driven shifts rather than abrupt fluctuations, which makes ETS suitable to model such deterministic trend behavior.

CPI Prediction (ARIMAX Model): The ARIMAX (Autoregressive Integrated Moving Average with Exogenous Variables) model is being used to forecast CPI values using tariff rates as an external variable, as seen in Equation 1. This method allows the model to account for both the autoregressive patterns with price indices and the direct influence of tariff changes. The

inclusion of tariff data is justified by our EDA, which revealed strong correlations between tariff rates and CPI movements across our selected industries. This provides a more causally interpretable framework than purely univariate time-series models.

Equation 1: ARIMAX Model to predict CPI values using Tariff Rates

$$\Delta CPI_t = c + \sum_{i=1}^p \phi_i CPI_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \beta * Tariff_t + \varepsilon_t$$

Where:

CPI_t is the Consumer Price index at time t

$Tariff_t$ is the tariff rate at time t (exogenous variable)

ϕ_i are the autoregressive coefficients

θ_j are the moving average coefficients

β is the impact of tariffs on CPI

ε_t is the random error term

p is the number of autoregressive terms & q is the number of moving average terms.

Interactive Visualization: Tableau forms the user-facing layer of the project, designed to make the modeling outputs accessible and visually intuitive. All cleaned datasets and ARIMA outputs were exported as structured CSV files. Tableau workbooks ingest these files and automatically refresh the visuals. We built the dashboard around four core visualization components:

1. Industry Metrics Explorer
 - a. Tariff, CPI, and PPI trends for electronics, apparel, and food.
 - b. Enables users to understand long-run movements and volatility.
2. Tariff–PPI and Tariff–CPI Relationship Trends
 - c. Time-series visuals highlight how Most-Favored Nation (MFN) tariff adjustments precede or coincide with changes in consumer and producer prices.
3. Global Tariff Map
 - d. A choropleth-style visualization showing tariff intensity across countries.
 - e. Helps contextualize U.S. tariff positioning globally.
4. Forecast-Based CPI Simulator
 - f. A novel interactive tool allowing users to select a month in 2025, an industry, and MFN tariff rate
 - g. Tableau then displays the predicted CPI based on the ARIMA forecast pipeline discussed above.
 - h. This simulator is the most innovative piece of the interface, bridging econometric outputs with user-driven scenario testing.

Evaluation

To assess both the analytical robustness of our models and the usability of our dashboard, we adopted a two-part evaluation framework. The first component focused on statistical model performance, while the second gathered user perspectives through a structured survey.

Evaluation - Algorithm

Our experiments were conducted to evaluate whether MFN tariff rates can meaningfully improve CPI forecasting accuracy across four categories: *Food, Apparel, Electronics, and Overall CPI* and to validate whether an ARIMAX-based simulator could reliably generate 2025 price forecasts under different tariff scenarios. All experiments followed a consistent pipeline: stationarity checks, transformation, parameter evaluation, model training/testing, and scenario forecasting.

Step 1: Tariff rates for each industry were used to fit an ETS model, with the “begin effect date” used to determine the applicable tariff rate for each industry. Because tariff adjustments were irregular, the data was split into an 80% training set and a 20% test set. ETS was also applied to forecast CPI values for key industries, using all data before 2024 for training and 2024 data for testing.

Observations: RMSE values for Food and Electronics were observed to be below 2%, and moderately higher for apparel at 5.7%. ETS provides an accurate baseline method for forecasting tariff rates. However, accuracy is limited by irregular adjustments where rates are not stable and do not follow a gradual trend. Furthermore, it does not account for exogenous effects, such as tariff rates, and was therefore not used to inform the dashboard.

Step 2: Stationarity Tests and Data Transformation to determine whether CPI datasets are stationary and identify necessary transformations prior to ARIMAX modeling. We began by testing all four CPI datasets for stationarity using the ADF test, which confirmed that each series was non-stationary in its raw form. To better understand the degree of persistence and trend components, we examined the corresponding ACF plots, which showed slow decay and strong autocorrelation. Based on these results, we applied log transformations followed by either first or second differencing (depending on the industry) to successfully convert each CPI series into a stationary form suitable for ARIMAX modeling.

Observations: All four CPI datasets were non-stationary in their raw form, with ADF tests showing high p-values and ACF plots indicating strong autocorrelation. Applying log transformation and differencing successfully produced stationary series across industries. These validated transformations enabled reliable ARIMAX modeling.

Step 3: Testing Exogenous Variable Relevance to confirm that tariff rates had statistically meaningful relationships with CPI changes.

Observations: All three industries showed positive correlations with tariff rates, with overall CPI exhibiting a moderate relationship that reflects broader inflationary pressures. These patterns supported the inclusion of tariffs as exogenous variables in the ARIMAX models.

Step 4: Model Order Selection (p, d, q) to identify the optimal ARIMAX parameters for each industry. We tested multiple combinations of ARIMA (p,d,q) parameters for each CPI series and evaluated their performance using the Bayesian Information Criterion (BIC). The model configuration with the lowest BIC value was selected for each industry, ensuring the best balance between fit and simplicity.

Final Selected Models (Lowest BIC):

- **CPI Apparel:** ARIMA (0, 1, 2)
- **CPI Food:** ARIMA (0,2,3)
- **CPI Electronics:** ARIMA (5, 2, 2)
- **CPI Overall:** ARIMA (0, 2, 4)

Step 5: Train–Test Evaluation on 2024 Data to assess predictive accuracy on unseen data and evaluate the forecasting reliability of each ARIMAX model. We evaluated model performance by splitting each dataset into a training set (all data before 2024) and a test set (2024 only). Using the trained ARIMAX models, we forecasted CPI values for 2024 and compared them to the actual observations, calculating MAPE and RMSE to quantify forecasting accuracy.

Observations: All four ARIMAX models closely tracked the actual 2024 CPI trends, with apparel showing the lowest error due to its stronger tariff sensitivity. Electronics exhibited slightly higher MAPE, reflecting greater global supply chain volatility. RMSE and MAPE for forecasted values using both ETS and ARIMAX are summarized in Table 1. ARIMAX performs better for apparel, while ETS performs better for food and electronics. With these mixed results, and no exogenous effects of tariff rates when

using ETS, ARIMAX was chosen for CPI prediction in the dashboard. Overall, the results confirmed that ARIMAX was suitable for forward-looking simulations.

Table 1: Evaluation Metrics for ETS and ARIMAX models used for CPI Forecasting

	ETS		ARIMAX	
Industry	RMSE	MAPE	RMSE	MAPE
Apparel	1.48	1.02	0.78	0.50
Food	0.49	0.14	3.41	0.73
Electronics	1.10	0.75	3.86	2.53

Step 6: Retraining Models for Scenario Simulation to build a forecasting engine for user-driven tariff scenarios. We retrained each industry-specific ARIMAX model using all available data from 2015 to 2024, then built a simulation framework that allows users to select an industry, choose a month in 2025, and input a hypothetical tariff rate. The model updates the exogenous variable accordingly and generates the corresponding adjusted CPI forecast.

Observations: Predicted CPI responded to tariff changes in a balanced way — sensitive but not overly reactive — consistent with economic intuition. Electronics showed the most muted pass-through, while food exhibited a quicker CPI response due to faster turnover. Apparel demonstrated the strongest marginal impact per tariff increase.

Step 7: Exporting forecasting output to CSV, connected them as data sources in Tableau, and built an interface featuring dropdowns for date and industry along with a tariff-rate slider.

Observations: The CPI curves aligned well with the raw historical trends, confirming that all data transformations were successfully reversed. Additionally, the simulation outputs showed continuous and monotonic CPI patterns, indicating that the integration of modeled values into the historical series was performed correctly.

Evaluation - User Feedback

After completing the dashboard, we published it to Tableau Public ([Tariff Impact Dashboard](#)) so users could easily access, test, and interact with the tool. The goal was to gather real user feedback and ensure the dashboard truly supports their tariff-related analysis needs.

To evaluate both the qualitative and quantitative benefits, we designed a structured user survey ([Tariff Impact Dashboard Survey](#)) using Microsoft Forms. The survey focused on evaluating user satisfaction, ease of use, and opportunities for improvement. We then shared the survey with users who are interested in the tariff topic and successfully collected 18 responses that provide meaningful insights for future enhancements.

Observations:

- Total Views on Published Tariff Impact Dashboard: 130 Views
- Survey Result: We collected 18 responses (Tariff Impact Dashboard User Survey__CSE 6264 - Team 107.xlsx).
 - Quantitative: Overall, the feedback shows strong quantitative results, with an overall satisfaction score of 4.56, ease of navigation at 4.61, clarity of visualizations also 4.61, the helpfulness of the tariff simulator at 4.56, forecast output reliability at 4.50, and understanding of tariff impact rated at 4.44, indicating that users generally found the tool intuitive, clear, and effective in supporting their analysis.
 - Qualitative: Users particularly appreciate the heat map, interactive features, and the tariff simulator, describing them as intuitive and helpful for understanding tariff impact. Qualitative responses highlight strengths in visual clarity and interactivity.

Conclusions and Discussion

This project produced a complete framework for analyzing how US tariff changes affect consumer prices across three industries. By collecting and cleaning 10 years of CPI and tariff data, we built ETS and ARIMAX forecasting models and integrated them into an interactive Tableau dashboard that enables users to visualize historical trends and simulate the inflationary impact of hypothetical tariff changes in the future. User feedback, collected through surveys, showed strong satisfaction with the clarity and usefulness of the dashboard, confirming that it is a good tool that shows tariff-CPI relationships to the public.

During model development, we encountered several limitations which shaped our final approach. Since datasets were highly non-stationary, they required log transformations and differencing before an ARIMAX model could be applied. However, the inverse log transformations to forecast CPI values caused exponential inflation and produced unusable confidence intervals, especially for tariff rates different from the historical rates. Scaling the tariff inputs reduced this effect, however it weakened the tariff pass through representation and a zero-value tariff became invalid. Additionally, tariff data was only available annually while CPI data was monthly, which led us to assume constant tariffs across each year, which may not reflect real policy dynamics. Despite these challenges, the project demonstrated the feasibility of linking tariff changes to CPI forecasts in an interactive visualization environment.

User feedback highlighted opportunities to further improve the Tableau dashboard usability. Some found the country filter confusing, the layout slightly difficult to navigate, and simulator results a bit hard to interpret. Enhancements such as simplifying layout flow, improving filter usability, adding tooltips or brief explanations to clarify forecast outputs, and refining color contrasts to further strengthen readability would improve the overall clarity and user experience of the dashboard. Future work may also incorporate more advanced forecasting models such as VARX, Prophet or Silverkite as these models can better handle non-stationarity, nonlinear relationships and irregular exogenous inputs and thus further improve the accuracy and stability of forecasting.

The final version of our interactive dashboard that incorporates our modeling outputs and design elements is shown in Figure 1 below.

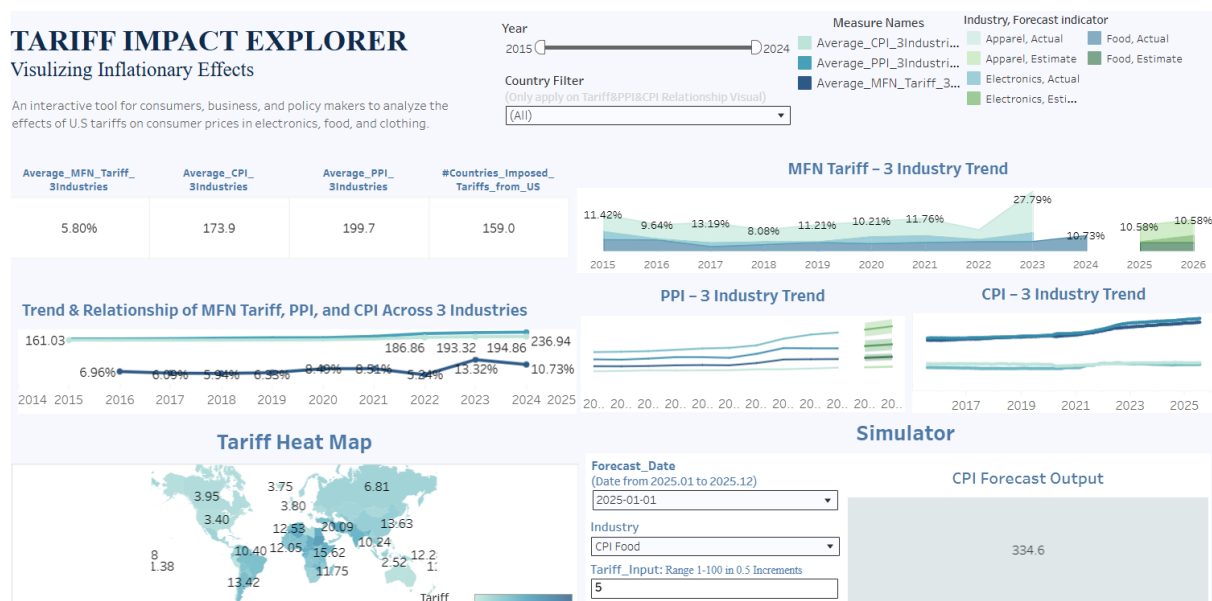


Figure 1: Final Version of Interactive Tableau Dashboard

All team members have contributed a similar amount of effort based on the assigned tasks.

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